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For this project I chose logistic topic on car parts as I am interested in automotive. These sample data that I have chosen are the things that usually happen to be the values used in automotive logistics. This variable might not be 100% accurate, but these are the main things that are usually used on transaction between the supplier and warehouse.

Sample Data

WarehouseID	Warehouse Location	Supplier Name	Contact	PartID	Part Name	Part Description	Price	OrderID	OrderDate	Quantity
W001	Jakarta	Supplier A	123456	P001	Engine Oil	Syntethic Oil	100	OR001	01/12/2024	50
W001	Jakarta	Supplier B	654321	P002	Brake Pad	Disc Brake Pad	200	OR002	03/12/2024	20
W002	Bandung	Supplier A	123456	P001	Engine Oil	Syntethic Oil	100	OR003	17/12/2024	30
W003	Surabaya	Supplier C	456321	P003	Air Filter	High-Quality Air	150	OR004	05/12/2024	10
W002	Bandung	Supplier B	654321	P002	Brake Pad	Disc Brake Pad	200	OR005	10/12/2024	25

This is the sample data that I am using which is also in the Un-Normal Form. This sample data represents the transaction between the warehouse and the supplier in which the information of the warehouse, supplier name, contact, parts, price, order details, etc. are stated in this table.

Anomalies in this UNF :

1. Cannot add new part without an order
2. Deleting order will remove detail about the warehouse and part
3. Changing the price of a part will need to update multiple row
4. There are repetition on supplier and warehouse information

2NF & 3NF			
Supplier			
SupplierID	Name	Contact	
S001	Supplier A	123456	
S002	Supplier B	654321	
S003	Supplier C	456321	
Warehouse			
WarehouseID	Warehouse Location		
W001	Jakarta		
W002	Bandung		
W003	Surabaya		

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The diagram above is the ERD that I have made for this project,

Car Part

Store information about the car parts.

- PartID **PK** : Unique ID that will represents each part.
- Name : Name of the specific part.
- Description : Description of the specific part.
- Price : Price of the specific part.

Supplier

Store information about the supplier of the product.

- SupplierID **PK** : Unique ID that represents the supplier of the product.
- Name : Name of the supplier.
- Contact : Contact details of the supplier.

Warehouse

Store information about the warehouse.

- WarehouseID **PK** : Unique ID that represents the warehouse.
- Warehouse Location : Location of the warehouse.

Order

Store information about the orders made by a supplier to a warehouse.

- OrderID **PK** : Unique ID that represents the order.
- SupplierID **FK** : Unique ID that connects to the supplier table to identify other information.
- WarehouseID **FK** : Unique ID that connects to the warehouse table to identify other information.
- OrderDate : Date of the order made.

OrderDetails

Store information about the order details.

- DetailID **PK** : Unique ID that represents the detail.
- OrderID **FK** : Unique ID that connects to the order table to identify other information.
- PartID **FK** : Unique ID that connects to the car part table to identify other information.
- Quantity : Quantity of part ordered.

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CREATE TABLE Supplier (

SupplierID VARCHAR(10) PRIMARY KEY,

Name VARCHAR(100),

Contact VARCHAR(15)

);

CREATE TABLE Warehouse (

WarehouseID VARCHAR(10) PRIMARY KEY,

Location VARCHAR(100)

);

CREATE TABLE CarPart (

PartID VARCHAR(10) PRIMARY KEY,

Name VARCHAR(100),

Description VARCHAR(255),

Price DECIMAL(10, 2)

);

CREATE TABLE Orders (

OrderID VARCHAR(10) PRIMARY KEY,

OrderDate DATE,

SupplierID VARCHAR(10),

WarehouseID VARCHAR(10),

FOREIGN KEY (SupplierID) REFERENCES Supplier(SupplierID),

FOREIGN KEY (WarehouseID) REFERENCES Warehouse(WarehouseID)

);

CREATE TABLE OrderDetails (

DetailID VARCHAR(10) PRIMARY KEY,

OrderID VARCHAR(10),

PartID VARCHAR(10),

Quantity INT,

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FOREIGN KEY (OrderID) REFERENCES Orders(OrderID),

FOREIGN KEY (PartID) REFERENCES CarPart(PartID)

);

INSERT INTO Supplier (SupplierID, Name, Contact)

VALUES

('S001', 'Alpha', '123456'),

('S002', 'Beta', '654321'),

('S003', 'Charlie', '456321'),

('S004', 'Delta', '789456'),

('S005', 'Eco', '147852');

INSERT INTO Warehouse (WarehouseID, Location)

VALUES

('W001', 'Jakarta'),

('W002', 'Bandung'),

('W003', 'Surabaya'),

('W004', 'Yogyakarta'),

('W005', 'Medan');

INSERT INTO CarPart (PartID, Name, Description, Price)

VALUES

('P001', 'Engine Oil', 'Synthetic Oil', 100.000),

('P002', 'Brake Pad', 'Disc Brake Pad', 200.000),

('P003', 'Air Filter', 'High-Quality Air', 150.000),

('P004', 'Spark Plug', 'High-Performance', 80.000),

('P005', 'Battery', 'Maintenance-Free', 300.000);

INSERT INTO Orders (OrderID, OrderDate, SupplierID, WarehouseID)

VALUES

('OR001', '2024-01-01', 'S001', 'W001'),

('OR002', '2024-03-12', 'S002', 'W001'),

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('OR003', '2024-12-17', 'S001', 'W002'),

('OR004', '2024-05-12', 'S003', 'W003'),

('OR005', '2024-10-10', 'S002', 'W002');

INSERT INTO OrderDetails (DetailID, OrderID, PartID, Quantity)

VALUES

('D001', 'OR001', 'P001', 50),

('D002', 'OR002', 'P002', 20),

('D003', 'OR003', 'P001', 30),

('D004', 'OR004', 'P003', 10),

('D005', 'OR005', 'P002', 25);